A Seminar

On

“Agentic AI: Autonomous Goal-Oriented Intelligent Systems for Planning, Reasoning and Task Execution”

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Abstract:

Agentic AI refers to AI systems that go beyond single-turn generation to autonomously pursue a user-specified goal by planning, invoking tools, observing outcomes, and adapting their next action, without step-by-step human instruction for every action [1]. Unlike conventional rule-based automation, generative AI, and standalone large language models (LLMs), which respond to a single prompt or a fixed pipeline, an agentic system decomposes a goal into subtasks, retrieves relevant knowledge, selects and invokes tools or APIs, and verifies its own results before proceeding [1][2][3]. Prior work has separately advanced planning and reasoning-action interleaving (ReAct) [4], memory and retrieval-augmented generation (RAG) [7], and multi-agent workflow coordination [5]; a parallel body of work catalogs the security, privacy, and ethical risks that arise once such agents act autonomously [6]. However, these two threads are rarely integrated into a single, demonstrable control loop: most reported architectures optimize for task capability, while risk assessment and outcome verification are treated as afterthoughts rather than first-class stages of the loop. This seminar surveys the core building blocks of Agentic AI — goal understanding, task decomposition, planning, memory/RAG, tool use, observation, and verification — and proposes a reliability-oriented agent architecture that explicitly separates planning from execution and pre-action risk screening from post-action verification, with bounded human-in-the-loop approval for high-risk actions and a retry/escalation policy bounded by a maximum retry count. The seminar evaluates this proposal qualitatively against a structural checklist of correctness rather than fabricated experimental results, and outlines a small prototype that demonstrates the loop end to end on safe, non-destructive tools.

Keywords: Agentic AI, autonomous agents, large language models, task planning, task decomposition, retrieval-augmented generation, tool use, reasoning and acting, human-in-the-loop, verification, risk assessment, multi-agent systems

1. Introduction

Artificial intelligence systems relevant to this seminar have evolved through three broad stages: (a) rule-based / conventional AI, which executes fixed logic with no learning or adaptation; (b) generative AI and large language models (LLMs), which produce fluent text, code, or other content in response to a prompt but do not, by themselves, act in the world or pursue a goal across multiple steps; and (c) Agentic AI, which uses an LLM (or a similar reasoning engine) as a controller inside a loop that plans, retrieves knowledge, selects tools, executes actions, observes outcomes, and

decides whether to continue, retry, or stop [1][2][3]. The distinguishing property of an agentic system is autonomous, goal-directed, multi-step behavior: given a goal, the system determines the sequence of sub-actions itself, rather than being told each step individually. This distinction matters practically — since an agentic system can attempt to actually perform actions rather than merely describe them, reliability, safety, and verification are central design concerns, not optional extras.

1.1. Motivation:

- LLMs are increasingly connected to real tools, APIs, files, and services, so a single incorrect or unverified action has real consequences — motivating an architecture where risk is assessed before an action executes, not only inspected after the fact [6].
- Existing literature richly covers individual capabilities — planning, ReAct-style reasoning [4], memory/RAG [7], and multi-agent coordination [5] — but rarely packages them into one teachable, reliability-first reference loop that a student or engineer can actually build and demonstrate.
- T.E. IT students entering industry will encounter agent frameworks and tool-calling copilots as part of everyday tooling; understanding the underlying control loop is a foundational, transferable skill.

1.2. Objective:

- Define Agentic AI precisely and distinguish it from conventional AI, generative AI, and standalone LLMs.
- Survey the core components of an agentic loop — planning, task decomposition, memory/RAG, tool use, observation, and verification — grounded in verifiable literature [1]-[7].
- Identify a concrete research gap in how existing work integrates capability with reliability governance.
- Propose a reliability-oriented Agentic AI architecture that separates planning from execution and pre-action risk screening from post-action verification, with bounded human-in-the-loop approval.
- Demonstrate the proposed control loop end-to-end through a small, safe, non-destructive prototype.

1.3. Problem Statement:

Existing research treats planning, memory and retrieval, tool use, multi-agent collaboration, and security/privacy risk as largely separate concerns, addressed in isolation across different surveys and systems [1][2][3][5][6]. A practical gap remains in integrating these capabilities into a minimal, teachable reference architecture that combines pre-action risk assessment, bounded human oversight, and post-action verification in one demonstrable control loop, in a way that a T.E.-level student can understand, diagram, and implement as a small working prototype. This seminar addresses that gap by proposing such an architecture and validating its feasibility through

a structural evaluation framework and a demonstrable prototype, rather than isolated capability benchmarks.

2. Methodology / Prototype / Idea / Algorithm

- Goal Understanding: interpret user intent and constraints from the stated goal.
- Task Decomposition & Planning: break the goal into ordered subtasks and select the next action using a ReAct-style reasoning-acting loop [4].
- Knowledge Grounding: retrieve relevant context via Memory + RAG before each action [7].
- Pre-Action Risk Gate: classify each action's risk and require human approval for high-risk actions before they execute [6].
- Post-Action Verification: observe the result, verify it against the expected outcome, and replan/retry within a bounded MAX_RETRIES before escalating to a human.

Figure 2.1 shows the resulting layered architecture: user goal at the input layer; reasoning, planning and task decomposition; knowledge grounding through Memory + RAG; a safety layer holding the risk gate and the human-approval step; and an execution/verification layer that feeds a bounded replanning loop back into the planner on verification failure.

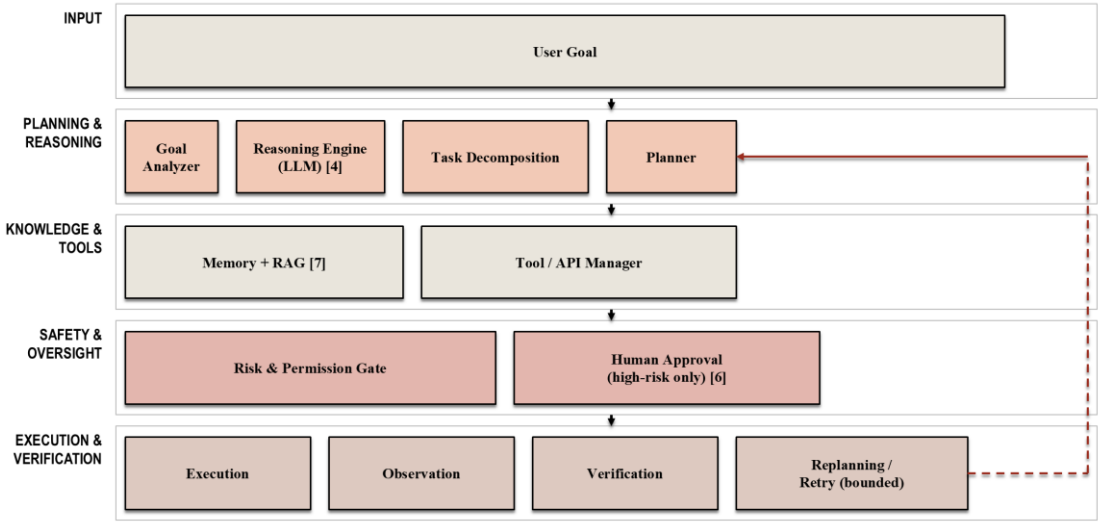


Figure 2.1: Proposed reliability-oriented Agentic AI architecture. The loop concludes with aggregated, verified outputs returned to the user after the Execution & Verification layer. Source: Author's proposed architecture, synthesized from [1], [2], [4], [7] and the risk taxonomy in [6].

Expected Outcomes:

This seminar is expected to: (i) define Agentic AI clearly relative to conventional and generative AI; (ii) synthesize a literature-grounded set of core agentic-loop components; (iii) identify and articulate a concrete research gap around reliability governance; (iv) propose a reliability-oriented architecture and bounded algorithm that separate planning from execution and pre-action risk screening from post-action verification; and (v) demonstrate the feasibility of that architecture

through a small, safe conceptual prototype (Phase 3), evaluated against the structural metrics in the evaluation framework rather than fabricated performance numbers.

Applications areas, if any:

Autonomous coding assistants that plan multi-file changes, run tests, and verify results before presenting a diff; enterprise workflow copilots that read documents and draft actions with human approval for customer-facing steps; research assistants that retrieve literature via RAG and flag claims needing human verification; and personal task agents where irreversible actions (payments, sending messages) are always gated behind explicit user approval.

3. Conclusion

Agentic AI extends LLMs from single-turn generation to autonomous, multi-step, goal-directed behavior by combining planning, memory/retrieval, tool use, and verification [1][2][3][4][7]. This seminar identified a gap in how existing work integrates that capability with reliability governance, and proposed an architecture that explicitly separates planning from execution and pre-action risk screening from post-action verification, with bounded human-in-the-loop approval and a retry-bounded verification loop. Future work includes completing the Phase 3 prototype, extending the architecture to multi-agent settings [5] while preserving the same reliability guarantees, and empirically evaluating the proposed metrics once a working system exists.

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